Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2016 JC2 Preliminary Examinations

H2 BIOLOGY **9648/2**

Paper 2: Core (Mark Scheme)

Monday 29th August 2016 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)
Answer **all eight** questions.
Write your answers in the spaces provided on the question paper.

Section B (Essay Question)
Answer **one** essay question only.
Write your answers on the separate answer paper provided.
All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,
1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/9
3	/13
4	/7
5	/11
6	/17
7	/11
Total	/80

This document consists of **19** printed pages.

[Turn over

Section A

Answer **all** *questions*.

1. Lysosomes play an important role in autophagy as they contain enzyme complexes that can break down worn out organelles. A section of the lysosomal membrane is magnified and shown in Fig. 1.1.

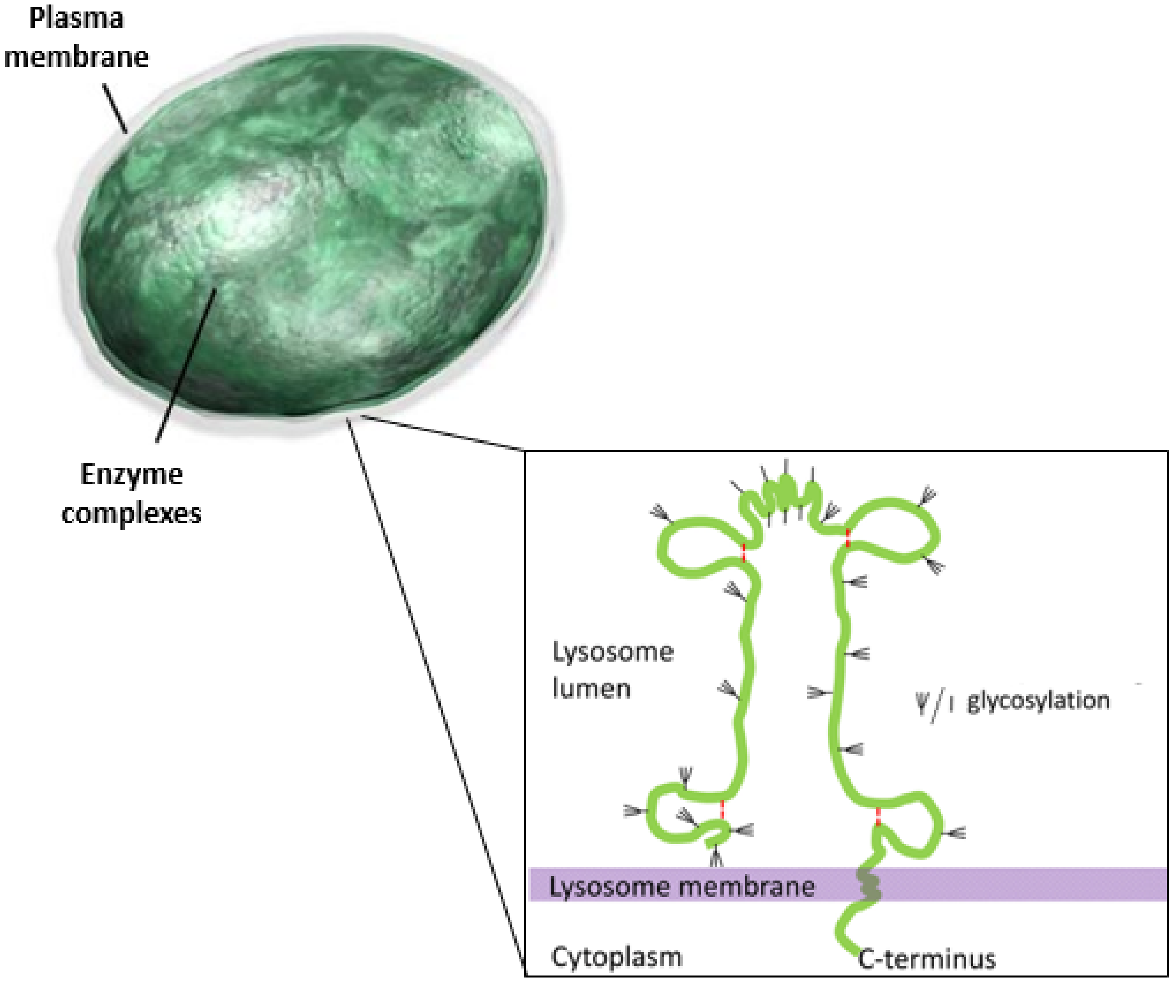


Fig. 1.1

(a) Describe and explain how the structure of the lysosomal membrane protects it from being degraded by the hydrolytic enzyme complexes it contains.

……………………………………………………………………………………………………[2]

1 Glycoproteins are embedded in the lysosomal membrane (at the C-terminus), **towards the lysosome lumen.**

2 This changes the **3D conformation** of the inner surface of the membrane, **not complementary/cannot recognise and bind to active sites** of hydrolytic enzymes**/no enzyme-substrate complexes** formed (hence, cannot degrade the lysosomal membrane).

(b) Collagen is an important structural component of skin, ligament and tendons. Describe the various levels of protein folding that gives it its fibrous structure.

………[4]

1 The primary structure of collagen is made up of **more than 1000 amino acid residues** and is **rich in glycines, lysines and prolines** / has a repeating motif of **Gly-X-(hydroxyl)Pro**.
2 In the secondary structure, each polypeptide folds spontaneously into a (alpha chain with a) **helical** structure.
3 In the quaternary structure, three polypeptides coil around each other to form a triple helix, held together by hydrogen bonds between –NH group of glycine on each strand and –CO group of amino acid residues on the other two strands; also involving hydroxyproline;
4 (Higher levels of structure) Multiple tropocollagen assemble in a staggered conformation, held together by covalent cross linkages between lysines and hydroxylysines (to form fibrils)
5 Aggregation of fibrils to form fibres.

(c) Fig. 1.2 and 1.3 show the structures of lysine and glycine respectively. In the space provided, draw a lys-gly dipeptide and label the peptide bond.

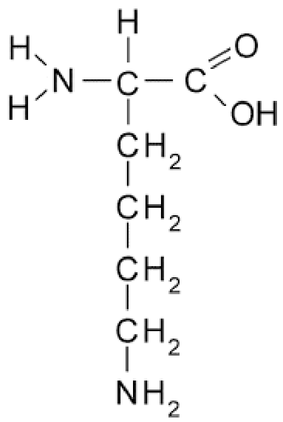


Fig. 1.2

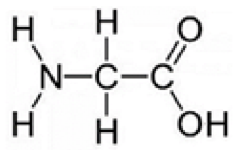
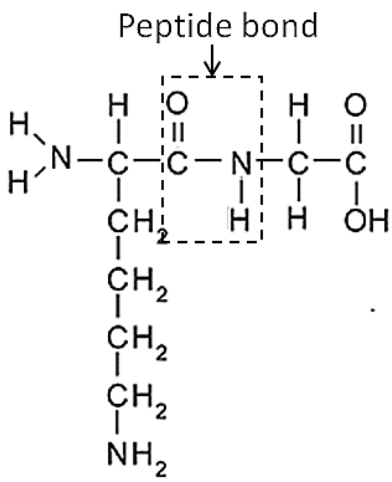


Fig. 1.3

………[1]



Mark scheme:

1 Correct dipeptide (N terminus first) AND
2 Labelling of peptide bond

(d) Fig. 1.4 shows some onion cells undergoing mitosis.

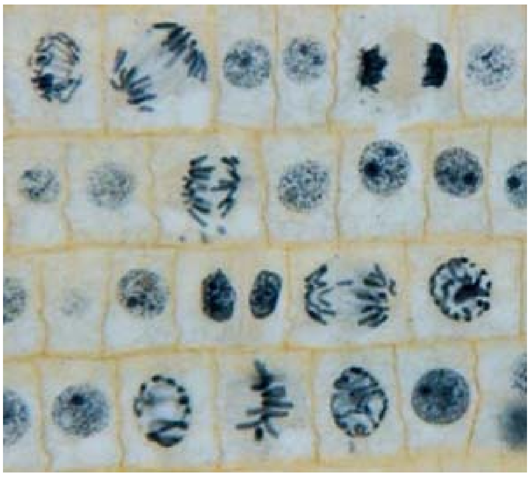
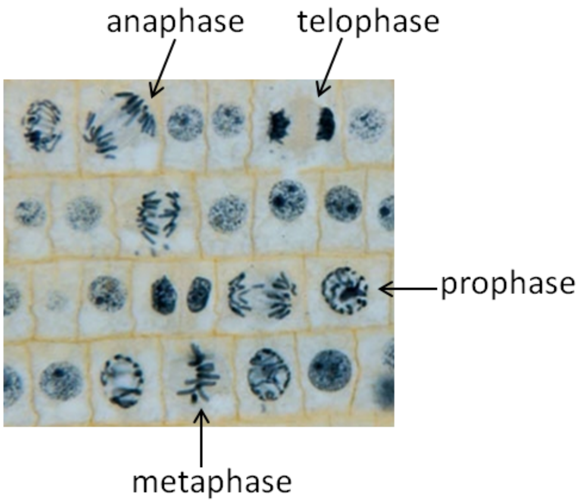


Fig. 1.4

(i) Identify (with arrows and labels) one cell each that is undergoing prophase, metaphase, anaphase and telophase.

……………………………………………………………..……………………….....……[1]



(ii) Describe the events of anaphase.

……………………………………………………………..……………………….....……[2]

1 <u>Centromeres divide</u> [Reject: centromeres split], sister chromatids separate into <u>individual chromosomes</u>
2 Sister chromatids are pulled to **opposite** **<u>poles</u>** of the cell by **shortening** of <u>kinetochore microtubules</u>

(iii) Describe the process of cytokinesis in the onion cells.

……………………………………………………………..………………………......……[2]

1 <u>Cellulose</u> (produced in the Golgi body) are packaged into <u>Golgi vesicles</u> and transported to the metaphase plate/middle of the cell;

2 Where they **fuse together** (and deposit the new cell wall materials), forming the <u>cell plate</u>, until it completely separates the two daughter cells.

[Q1 Total: 12]

2. Fig. 2.1 shows a process happening in a eukaryotic cell.

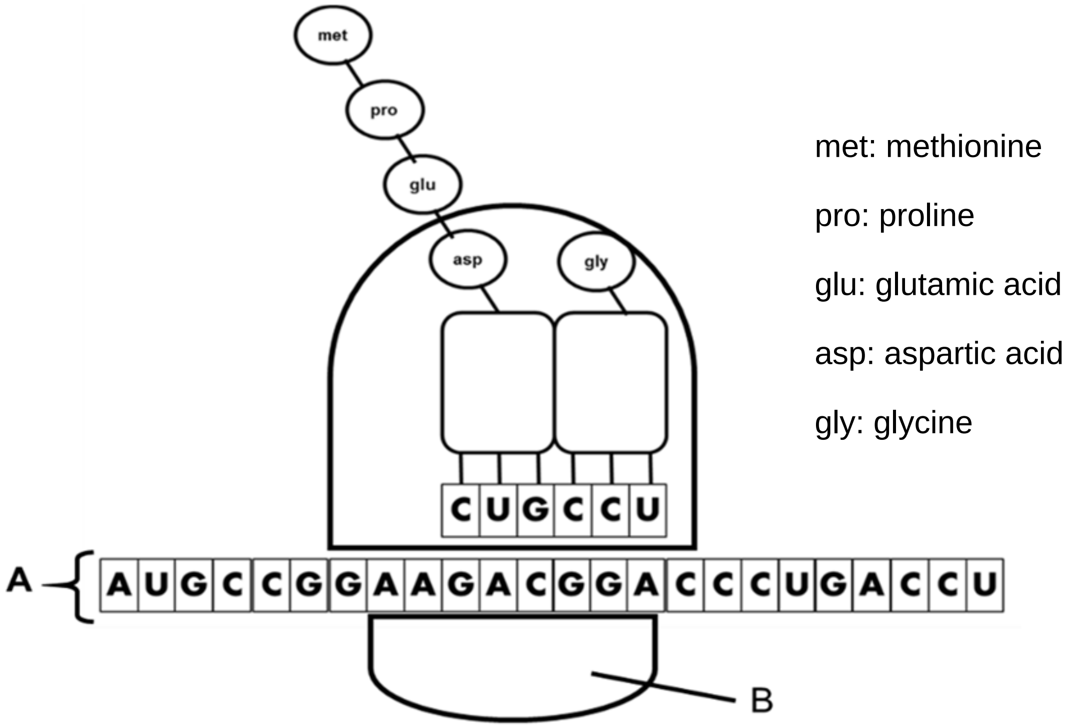


Fig. 2.1

(a) Name the structures labelled **A** and **B**. [1]

A: ________________________________

B: ________________________________

A: messenger RNA
B: small ribosomal subunit [Accept: ribosome]

(b) Relate the structure of **A** to its role in protein synthesis.

……[1]

1 Sequence of A is complementary to DNA template; hence, can relay information from nucleus from cytoplasm
/Codons in mRNA are complementary to/form complementary base pairs with anti-codons in tRNA; specify the sequence of amino acids in a polypeptide;

2 Start and stop codon; determines initiation and termination of translation;

(c) Outline one change to **A**, following completion of transcription in the nucleus before it can used for the process in Fig. 2.1.

……………………………………………………………………………………………..[1]

1 Splicing: <u>Excision of introns</u> and <u>ligation of exons together</u> by <u>spliceosomes</u> / Polyadenylation/addition of a **long chain of adenine nucleotides** to <u>3' ends</u> of the pre-mRNA.

(d) How many amino acids does the protein, encoded by the section of **A** shown in Fig. 2.1, have?

……………………………………………………………………………………………..[1]

1 6 (UGA is a stop codon)

(e) An antibiotic which affects the elongation stage of the process in Fig. 2.1, is added shortly after initiation, such that truncated polypeptides are formed instead.

With reference to Fig. 2.1, suggest and explain how the antibiotic works to give rise to a tripeptide.

……………………………………………………………………………………………..[3]

1 Binds to **A-site**/aminoacyl (tRNA binding) site on ribosome (by complementary shape),
2 Prevents the entry of aminoacyl tRNA carrying **aspartic acid**/activated **aspartic acid**;
/Prevents complementary base pairing between anti-codon, <u>CUG</u>, of tRNA (carrying aspartic acid) and corresponding mRNA codon <u>GAC;;</u>
3 No formation of <u>peptide bond</u> between **glutamic acid and aspartic acid**. (Elongation cannot proceed).

AVP:
1 Inhibits <u>peptidyl-transferase</u> of large ribosome;;
2 Ref. competitive/ non-competitive inhibition mechanisms;;
3 No formation of <u>peptide bond</u> between **glutamic acid and aspartic acid**. (Elongation cannot proceed).

(f) Tetracycline is an example of a ribosome-targeting antibiotic which is effective only towards bacterial cells.

Suggest why tetracycline has no effect on eukaryotes.

……………………………………………………………………………………………..[2]

1 Eukaryotes have 80S ribosomes but bacteria have 70S ribosomes;;
2 Tetracycline is only **complementary in shape** to the A-site/aminoacyl (tRNA binding) site/<u>large</u> ribosomal subunit of the 70S ribosomes/only able to enter the A site of <u>large</u> ribosomal subunit of 70S ribosome;;

[Q2 Total: 9]

3. A study published in the *Proceedings of the National Academy of Sciences* revealed a plausible strategy to facilitate the replacement of antibiotic resistant pathogens with sensitive ones in hospitals, which faced a higher rate of evolution of antibiotic resistance.

Researchers developed and administered two different phages that target the common gut microbe *E. coli*. The first was just a standard "lytic" phage that results in the death of the bacterial cell. The second phage, called a "temperate" phage, was a little more special. The researchers engineered it to possess a well-known gene-editing system that, when injected into the host cell, both removes their antibiotic resistance genes and renders them resistant to the lytic phage at the same time.

To select for bacteria that has been successfully edited by the temperate phage, the lytic phage is added. Lytic phages only target bacteria that has not undergone gene-editing, thus selecting for successfully edited bacteria that are now antibiotic-sensitive.

Both phages designed according to this strategy are used on hospital surfaces and hand sanitizers to restore a "healthy balance" in the population of antibiotic-sensitive bacteria and antibiotic-resistant bacteria.

(a)(i) Explain how the structure of a "lytic" phage facilitates the "death of the bacterial cell."

…..…………………………………...……………………..……………..……………..[2]

1 Viral **DNA genome** codes for **proteins**; which
hydrolyses host DNA
/ shuts down macromolecular (DNA, RNA, protein) synthesis
/ direct host's transcription and translation machinery, e.g. RNA polymerase and ribosomes, to synthesize phage enzymes and phage structural components for the assembly of new phages; [any 1]

2 (Phage-encoded) <u>lysozyme</u> which <u>lyses</u> the bacterial cell (to facilitate release of phages, thus, killing the cell);;

(ii) Suggest why it is important to restore a "healthy balance" in the population of antibiotic-sensitive bacteria and antibiotic-resistant bacteria in hospitals.

…..…………………………………...……………………..……………..……………..[1]

1 To slow down the rate of evolution of antibiotic resistance in bacteria/allow less antibiotic resistant bacteria to be present; so that the antibiotics in use currently are still effective;

(iii) Contrast the life cycle of the "temperate" phage and life cycle of a HIV.
……………………………………………………………………………………………………..[1]

	Temperate phage	HIV
1	Host cell is *E.coli*.	Host cell is T lymphocytes/ macrophages
2	Attachment sites on tail fibre recognize and bind to specific receptors on host cell	Gp120 recognise and bind to CD4 receptor on host cell
3	Only <u>genome enters</u> cytoplasm	Entire viral <u>nucleocapsid enters</u> cytoplasm
4	<u>Injection</u> of genome via the hollow tube	<u>Fusion</u> of membrane envelope with host cell membrane.
5	No uncoating of nucleocapsid in cytoplasm.	Uncoating occurs in cytoplasm.
6	No reverse transcription occurring (by reverse transcriptase) / viral DNA integrates into host genome **directly**	Viral RNA undergoes <u>reverse transcription</u> to form viral cDNA (by reverse transcriptase).
7	Upon activation/induction, <u>excision</u> of prophage from host genome.	Upon activation/induction, <u>no excision</u> of provirus from host genome.
8	Virus can switch to <u>lytic mode</u>	Virus will not switch to lytic mode.
9	During maturation, there is incorporation <u>of lysozyme</u> into the tail fibre.	During maturation, polyproteins are cleaved by host or viral <u>proteases</u> to yield <u>functional</u> viral proteins.
10	No insertion of viral glycoproteins into host cell membrane.	New viral glycoproteins (gp120 and gp41) are made and inserted into the host cell membrane.
11	New viral particles released from host cell after **lysis.**	New viral particles are released out of cell after <u>budding</u> of nucleocapsid from the host envelope (used as viral envelope)

[Any 1]

(b)(i) In farms, poultry are often infected with *E. coli*, some of which are resistant to antibiotics. An alternative to treating these poultry with antibiotics is to inject them with “lytic” phages. Explain how this treatment may result in the spread of antibiotic resistance in the *E.coli* that infects the poultry.

…..…………………………………..……………………..……………..……………..[4]

1 Phage undergoes <u>generalised transduction</u> (during the lytic cycle);;
2 **Donor** bacteria DNA is hydrolysed into fragments (by phage proteins);;
3 (Random) packaging of host **antibiotic resistance genes** into the capsid of newly assembled phages;
4 Phage released can **infect another** bacterium, transferring the antibiotic resistance genes via **genetic**/**homologous recombination / insertion / integration**;
5 **expression** of antibiotic resistance genes result in production of antibiotic resistance proteins in the newly infected cell (to give rise to the phenotype of antibiotic resistance);;

(ii) The normal life cycle of the “lytic” phage did not involve homologous recombination of viral DNA genome with the host cell genome.

Suggest why homologous recombination with the host cell genome is possible in generalised transduction.

…..…………………………………..……………………..……………..……………..[1]

1 **Bacterial/donor** antibiotic resistance genes transferred over by the “lytic” phage is **homologous** to the recipient cell’s genome;;

(c) In another study, a mutant *E. coli* strain is engineered to have the following genotype:

trpR$^+$_trpP$^+$_trpO$^-$_trpE$^+$_trpD$^+$_trpC$^+$_trpB$^+$_trpA$^+$

Explain how this bacteria strain will respond in the presence of tryptophan (+: wild type allele; **-** : loss of function allele).

…..…………………………………..……………………..……………..……………..[4]

1 tryptophan act as a <u>co-repressor</u>; which binds to the <u>allosteric site</u> of repressor;
2 changes the **3D conformation** (of the DNA binding site) of the repressor to make it **active**;
3 cannot bind to the mutant operator; because DNA binding site (of repressor) is no longer **complementary in shape** to the mutant operator sequence;
4 RNA polymerase continue to **access and transcribe the structural genes**; structural **gene products** are still produced;

[Q3 Total: 13]

4. Fig. 4.1 shows the relationship between telomere lengths and cell divisions in somatic cells and embryonic stem cells.

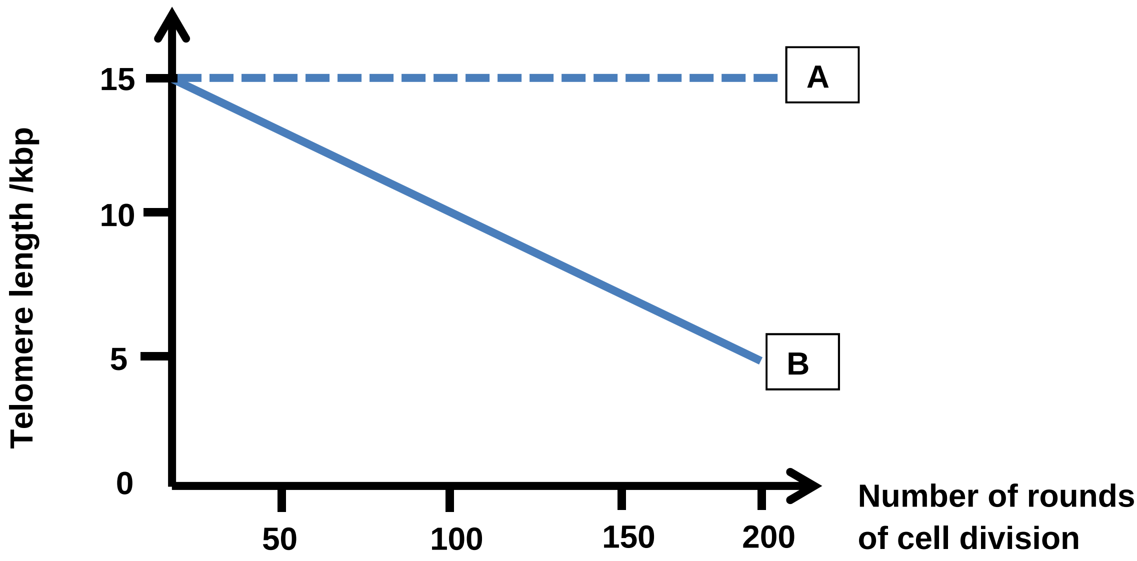


Fig. 4.1

(a) Identify the graph that represents the respective cell type. [1]

Somatic cells: Graph …………

Embryonic stem cells: Graph …………

1 Somatic cells: Graph ……**B**……
2 Embryonic stem cells: Graph ……**A**……

(b) Explain your choice in **(a)**.

……………………………………………………………………………………………………[4]

Somatic cells

1 As the number of cell division increases (progressive cell divisions) from 0 – 200 , telomere length decreases from 15 – 5 kbp
2 During DNA replication, when the last primer (on lagging strand) is removed, no free 3' OH available for DNA polymerase to add (deoxy)ribonucleotides to complete the synthesis of the 5' ends of **daughter** DNA strands (telomeres shorten with each round of replication)

Embryonic stem cells

1 As cell division increases (progressive cell divisions) from 0 – 200, telomere length remains constant at 15 kbp
2 Active telomerase present adds **telomeric** DNA sequence to the 3' ends of DNA **parental/template** strands (allowing DNA polymerase to complete the synthesis of the 5' end of daughter DNA strands)

(c) Describe a role of telomeres.

……………..…………………………………………………..……………………......……[1]

1 Serves as disposable buffer to protect the coding DNA from gene erosion during DNA replication as DNA shortens with each round of replication (end-replication problem)

2 Telomeric DNA bind proteins to form a protective nucleoprotein cap which serves the following functions [Any one function]:

a) Protects the end of the chromosome from degradation by nucleases.

b) Prevents end-joining of chromosome ends which may lead to chromosomal mutations

c) Preventing unintentional cell death

The regulation of gene expression in somatic cells of eukaryotes differs from that in prokaryotic cells.

(d) Describe one difference in their gene expression at the transcriptional level.

……………..…………………………………………………..……………………......……[1]

Prokaryotes	Eukaryotes
• DNA and histone modifications cannot occur as prokaryotic DNA not organized into chromatin / not associated with histones	• DNA and histone modifications can occur, resulting in **conversion between euchromatin and heterochromatin** (as eukaryotic DNA is complexed with histones and other proteins to form chromatin / associated with histones)
• Major on/off switch for gene regulation are DNA sequences, e.g. **promoters and operators**	• Major on/off switch for gene regulation is the structure of chromatin – **euchromatin, ready to be transcribed, or heterochromatin inaccessible to transcription enzymes**
• **Simple interactions** between transcriptional regulatory proteins and DNA-binding sites upstream of the cluster of structural genes to regulate **initiation** of transcription	• **Complex interactions** between protein factors and DNA sequences upstream of gene to **stimulate/repress and initiate** transcription
• Transcription of structural genes in the same operon is controlled by only **1 promoter**	• Transcription of each gene controlled by **own promoter** [Reject: eukaryotic genes are not clustered into operons ➔ needs more elaboration on idea of operon]

[Q4 Total: 7]

5. Coat colour in mice is determined by two gene loci, **B/b** and **D/d**.

The alleles give the following phenotypes:

B black **b** brown
D coloured **d** albino (white)

A cross between two black coat mice that are heterozygous at both gene loci produces F_1 offspring of the following phenotypes:

Black coat	675
Brown coat	225
Albino coat	300

(a) Define the term heterozygous.

……………………………………………………………………..……………….……....……[1]

1 Genotype of two **different** alleles at a (particular) gene <u>locus</u> of **homologous chromosomes**

(b) Draw a genetic diagram to show the cross described. [4]

Parental phenotypes : Black coat x Black coat

Parental genotypes (2n) : BbDd x BbDd

Parental gametes (n) BD Bd bD bd BD Bd bD bd

Punnett square to show F_1 generation genotypes:

	BD	Bd	bD	bd
BD	BBDD Black coat	BBDd Black coat	BbDD Black coat	BbDd Black coat
Bd	BBDd Black coat	BBdd Albino coat	BbDd Black coat	Bbdd Albino coat
bD	BbDD Black coat	BbDd Black coat	bbDD Brown coat	bbDd Brown coat
bd	BbDd Black coat	Bbdd Albino coat	bbDd Brown coat	bbdd Albino coat

F_1 genotypic ratio :	**9 B_D_**	**:**	**3 bbD_**	**:**	**4 _ _ dd**
F_1 phenotypic ratio:	**Black**	**:**	**Brown**	**:**	**Albino**
	9	**:**	**3**	**:**	**4**

Mark scheme:

1 Parental gametes – (Gametes **must** be circled)
2 F_1 genotypes / Punnett square
3 Genotypes correspond to phenotypes / legend for Punnett square
4 F_1 phenotypic ratio

(c) Explain how the interaction between the two genes B and D gives rise to albino coat colour.

…………………………………………………………………………………………………[4]

1 The dd genotype is <u>epistatic</u> over the B/b gene locus;
2 B/b locus code for <u>enzymes/proteins</u> involved in the production/synthesis of pigments;
3 D allele codes for <u>protein</u> D that deposits pigment/transports pigment to certain locations;
4 Genotype dd results in the **absence** of pigment depositor/transporter protein, resulting in no pigment deposition (regardless of the genotype at the *B/b* gene locus; thus, albino coat colour);

A wheat breeder crossed two cultivars of wheat that varied in kernel colour – one with dark red kernel, another with white kernel (a kernel is the seed from which the wheat plant grows)

He noted that the F_1 generation was intermediate in colour (light red), but the F_2 generation ranges in colours from white to dark red.

Fig. 5.1 shows the range of kernel colour in the F_2 generation.

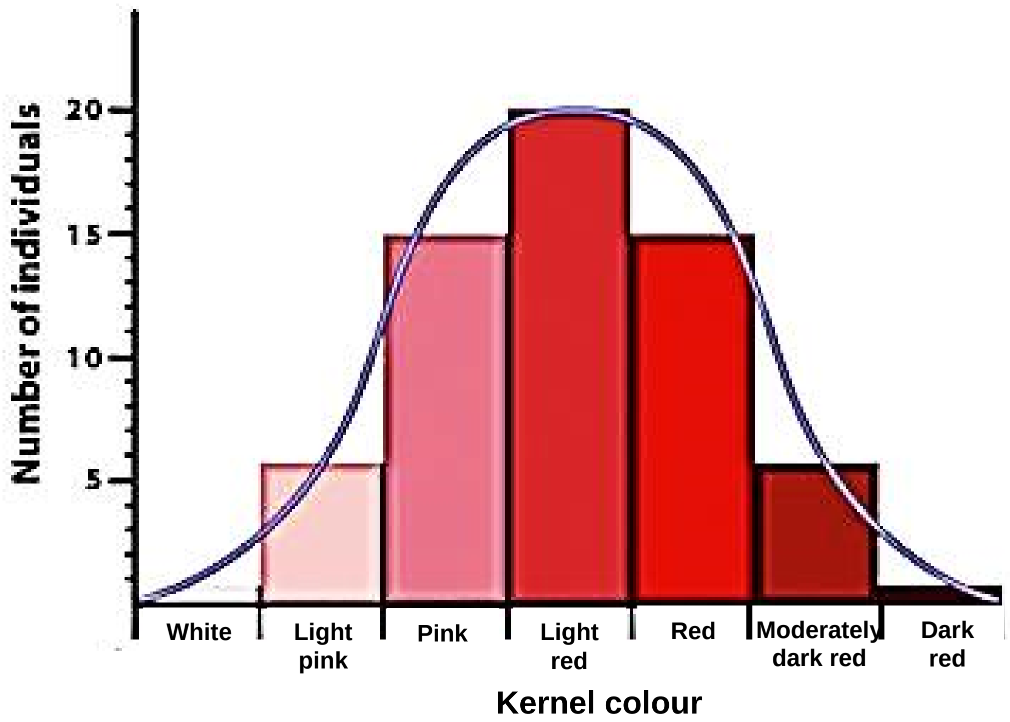


Fig. 5.1

(d) Explain the genetic basis behind the wide range of kernel colours.
…………………………………………………………………………………………………[2]
1 **Many** genes control the kernel colour (polygenic)
2 Each gene has an additive effect on the phenotype

[Q5 Total: 11]

6. It has been hypothesized that the neurotransmitter, serotonin, plays a part in appetite control. A consistent pattern of reduction in hunger motivation is seen in human studies with a variety of drugs which induce the release of serotonin.

Patients receiving these drugs report both a lower frequency and a reduced strength of urges to eat, together with the feeling of being more in control of their eating. Such serotonergic drugs, such as Dexfenfluramine, can exert a continued suppression of appetite to bring about subsequent weight loss.

Fig 6.1 shows the diagram of a synapse at which serotonin is the neurotransmitter.

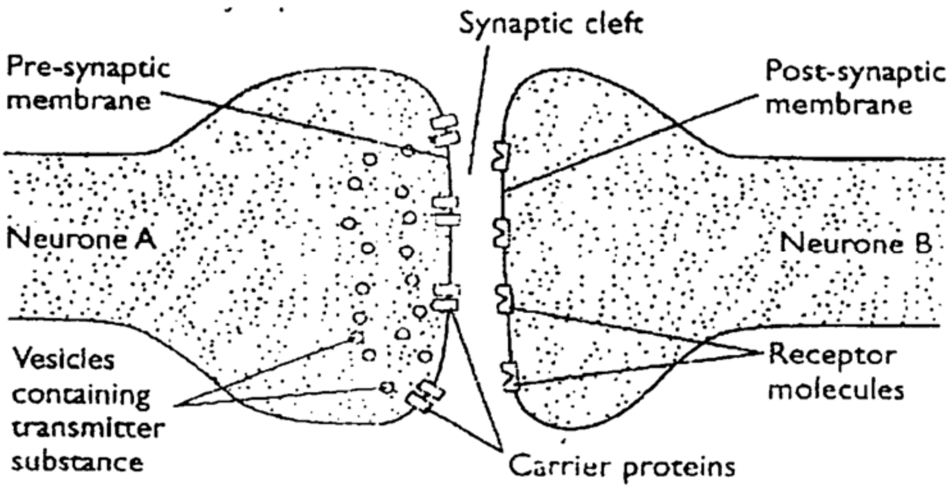


Fig 6.1

(a)(i) Describe how an impulse is initiated in Neurone B and transmitted to the brain, after the ingestion of Dexfenfluramine, to achieve the perception of fullness.

……………………………………………..…………………………………..……………..….[4]

1 (Dexfenfluramine induces the release of serotonin.) Serotonin diffuses across synaptic cleft and bind to specific receptor sites / ligand-gated Na^+ channels (on the post-synaptic membrane)

2 Ligand-gated Na^+ channels **open**; causing diffusion of Na^+ **into** post-synaptic neurone;

3 Post-synaptic membrane becomes depolarized; and excitatory post-synaptic potential (EPSP) is generated;

4 Summation of EPSPs at axon hillock; results in post-synaptic membrane reaching threshold potential of -50mV;
(Action potential is generated in post-synaptic membrane to initiate an impulse to the brain)

Serotonin is normally rapidly reabsorbed from the synaptic cleft by carrier proteins in the pre-synaptic membrane.

(ii) Suggest one advantage of rapidly reabsorbing serotonin.

…………………………………………..……………………………………………...….[1]

1 Prevents **continued** stimulation of the postsynaptic membrane;;
/ Allows transmission of impulse to be **rapidly** terminated;;

(iii) Explain the effect of mitochondrial poisons, which inhibit cytochromes/electron carriers, on synaptic transmission.

…………………………………………..……………………………………………...….[2]

1 No synthesis of ATP; due to no electron transport down the ETC/no proton gradient synthesis/no chemiosmosis;
2 No exocytosis of synaptic vesicles, involving the **movement** of synaptic vesicles towards the pre-synaptic membrane, (fusion with it and the release of serotonin);;

Serotonin receptors consist of several subtypes. In addition to ligand-gated ion channels, another subtype of serotonin receptors is shown in Fig 6.2, which participates in a transduction pathway that reduces the generation of beta amyloid, a compound involved in Alzheimer's disease. Raf, MEK and ERK are kinases.

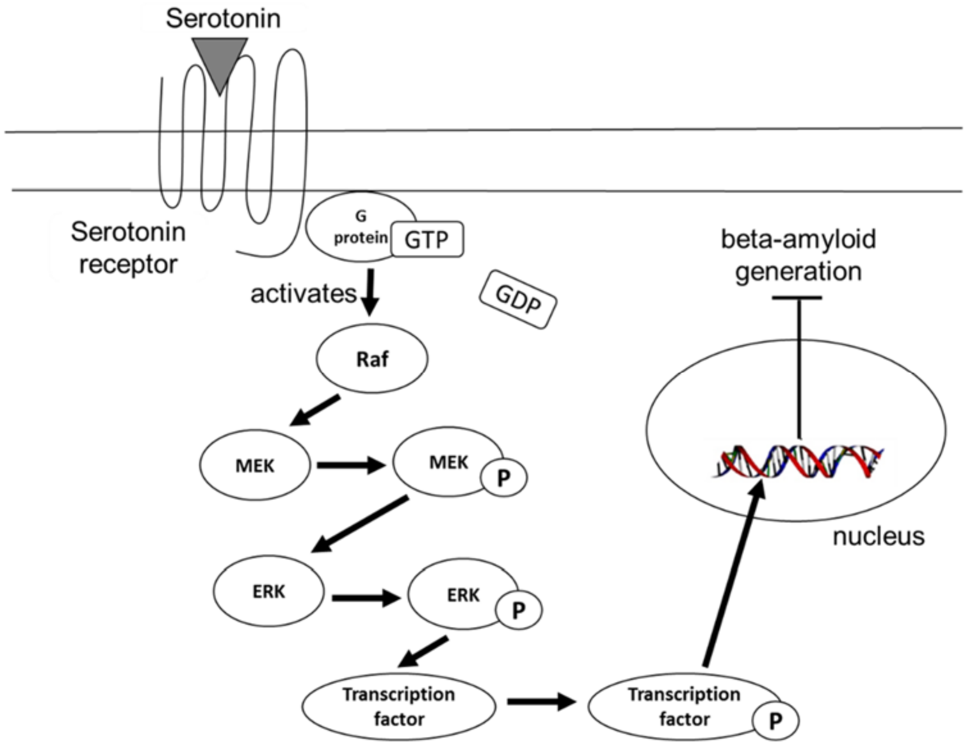


Fig. 6.2

(b)(i) Explain how the receptor in Fig.6.2 stays embedded in the plasma membrane and retains its mobility in the membrane.

……………………………………………………………………………………………..……[3]

1 <u>Non-polar / hydrophobic R groups / side chains</u> of amino acids of protein; Interact with <u>hydrophobic core / fatty acid tails</u> of membrane;

2 <u>Polar / charged / hydrophilic R groups / side chains</u> of amino acids of protein; interact with <u>hydrophilic phosphate head</u> of membrane;

3 Receptor is mobile in the membrane due to the membrane structure is **fluid;;**

(ii) With reference to Fig. 6.2, explain how the binding of serotonin leads to a reduction in beta-amyloid generation.

……………………………………………………………………………………………..……[4]

1 Binding of serotonin triggers **conformational** changes which exposes the **cytoplasmic** domain of the GPCR (receptor is activated);;

2 G-protein **binds** to GPCR and **exchanges** GDP for GTP; and is **activated**;;

3 G-protein activates <u>Raf</u>; activated Raf phosphorylates and **activates** <u>MEK</u>; Activated MEK phosphorylates and **activates** <u>ERK</u>;

4 Activated ERK phosphorylates and **activates** a <u>transcription factor</u>, which **enters the nucleus** and **binds to its DNA elements**;;

5 Ref. Differential gene expression results in reduction in / no generation of beta-amyloid;;

(c) There are reactions in plants which are crucial for its survival and growth. Explain the effect of a base addition to the gene coding for RuBisCO (Ribulose-1,5-bisphosphate carboxylase) enzyme.

…………………………………………………………………………………………..……[3]

1 Addition of a base leads to **frameshift** mutation; which results in a **different mRNA sequence**;
2 Different amino acid sequence / primary structure (of polypeptide);
Different R group (as original amino acid);
3 Different 3D configuration of enzyme
/ Different 3D shape of active site;;
4 No fixing of carbon dioxide to RuBP to form PGA (in Calvin cycle);;

OR

5 Addition of a base leads to **frameshift** mutation; which results in a **different mRNA sequence**;
6 Generation of **new stop codon**, resulting in **truncated** protein
7 No fixing of carbon dioxide to RuBP to form PGA (in Calvin cycle);;

[Q6 Total: 17]

7. Bartesian mimicry is a type of mimicry in which a harmless organism is protected by its resemblance to another species which is avoided by predators. An example of this is the king snake, which is quite harmless. However, it has a similar colour pattern as the coral snake which is highly venomous. As a result, predators of snakes that avoid the coral snake also avoid the king snake as well.

(a)(i) Suggest an evolutionary advantage to the king snake for merely resembling a venomous coral snake instead of producing true venom.

…………………………………………………………………………………………………….........…[1]

1 King snakes can save energy and resources (like amino acids) which would be needed to produce the venom and use them instead **for reproduction / foraging**.

(ii) Discuss how natural selection could have led to the evolution of the colour pattern in the king snake.

…………………………………………………………………………………………………….........…[4]

1 <u>Genetic variation</u> in colour pattern exists in king snake populations due to <u>mutations</u>.
2 King snakes are subjected to <u>selection pressure</u> + e.g. **predation**
3 Individuals with colour patterns that more closely resemble coral snakes are at a **selective advantage + reason** for avoidance: such snakes are thought to be venomous;
4 Such snakes can better **survive to reproductive age** and **reproduce**, **passing down the advantages alleles** [reject: traits/characteristics] to their offspring.
5 Over time, <u>allele frequencies</u> change and the colour pattern resembling coral snakes become **predominant**.

(b) DNA–DNA hybridisation is a technique used to determine the similarity of DNA from different species.

In this technique:

- DNA is extracted from four king snake species and cut into pieces using special enzymes;
- the DNA is heated to separate the double strands into single strands;
- single-stranded DNA from different pairs of king snake species are mixed together and cooled so that double strands of hybrid DNA form.

Fig. 7.1 shows the temperature (°C) needed to separate the hybrid DNA strands of the four king snake species, *L. getula, L. calligaster, L. triangulum and L. nigrita.*

Fig. 7.1

	L. getula	*L. calligaster*	*L. triangulum*	*L. nigrita*
L. getula	91	84	89	80
L. calligaster		90	86	87
L. triangulum			92	82
L. nigrita				90

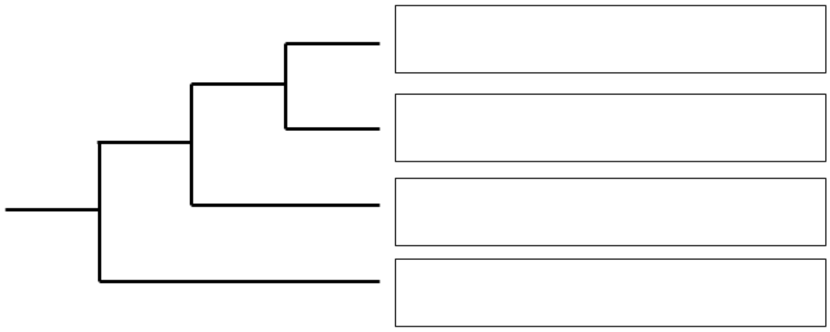
(i) Describe and explain the difference in denaturation temperature for the *L. getula–L. triangulum* hybrid DNA and the *L. getula–L. nigrita* hybrid DNA.

……………………………………………………………………………..……………....................….[3]

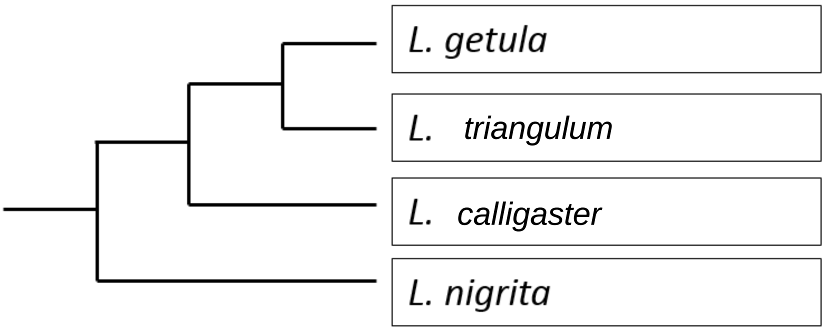
1 The denaturation temperature of *L. getula–L. triangulum* hybrid DNA of <u>89°C</u> is **higher** than that of the *L. getula–L. nigrita* hybrid DNA, which is <u>80°C.</u>
2 There is **higher molecular homology** between the *L. getula* and *L. triangulum* DNA than between *L. getula* and *L. nigrita* DNA.
3 Hence **more hydrogen bonds** are found in the *L. getula–L. triangulum* hybrid DNA than in the *L. getula–L. nigrita* hybrid DNA, hence requires higher temperature to denature.

(ii) Using the data in Fig. 7.1, fill in phylogenetic diagram below to show the evolutionary relationship between the four species of king snakes.

……………………………………………………………………………..……………....................….[1]



Ans:



(iii) Discuss how the neutral theory of evolution could be used to determine when two species of king snakes diverged from their common ancestor.

……….[2]

1 Neutral mutations do not confer any selective advantage or disadvantage **;** are fixed/accumulated at a relatively constant rate for any particular gene like a molecular clock,

2 By determining the number of mutations/substitutions/differences**;** in a common gene/amino acid sequence in two species of king snakes, we could calculate the length of time since divergence**;**

[Q7 Total: 11]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

8 **(a)** Explain the advantages and significance of cell signalling. [5]

(b) Secretion of glucagon plays a critical role the regulation of glucose concentration in the bloodstream.

With reference to the stages of cell signalling, describe how glucagon performs this role. [10]

(c) Discuss the differences between the cell signalling pathways of insulin and glucagon. [5]

[Total: 20]

OR

9 **(a)** Describe, using examples, how homology supports Darwin's theory of natural selection. [10]

(b) Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [10]

[Total: 20]

8

(a) Explain the advantages and significance of cell signalling. [5]

1 Signal amplification, where **small quantities of ligand** are sufficient to bring about a **very large cellular response**.

2 Interaction/cross-talk between different signaling pathways allows for **regulation or control of cellular response.**

3 Specificity of cell signaling, where the the specific interaction between a receptor and its complementary ligand **ensures the appropriate response from its target cell.**

4 **Hydrophilic ligands** are able to **activate gene expression in the nucleus** even though these molecules cannot traverse the **hydrophobic plasma membrane**. This allows for transmission of signal across membrane barriers.

5 Single signal molecule can trigger multiple responses. The **use of the same protein in more than one pathway** allows cells to **economize on the number of different proteins it must make.**

6 Activation of many cells simultaneously, where a single signal molecule can activate multiple cell/cell types to **give a coordinated response within body**.

7 Signals can be terminated to **prevent unintended prolonged cellular responses** which may be **harmful**.

(b) Secretion of glucagon plays a critical role the regulation of glucose concentration in the bloodstream.

With reference to the stages of cell signalling, describe how glucagon performs this role. [10]

Ligand-receptor interaction

1 The ligand glucagon binds to a G-protein-coupled receptor (GPCR) located on the cell surface membrane of the liver/target cell;
2 Triggers a conformational change in receptor which exposes the cytoplasmic domain of GPCR (activating the receptor)

Signal Transduction

3 Inactive G protein binds to receptor (cytoplasmic domain), exchanges GDP for GTP and becomes activated;
4 (Activated) G protein dissociates from the receptor, diffuses along the plasma membrane, binds to and activates the enzyme adenylyl cyclase;

5 Adenylyl cyclase catalyses the conversion of ATP to cyclic AMP (cAMP), which acts as a second messenger by activating protein kinase A;
6 protein kinase A then activates phosphorylase kinase via phosphorylation,
7 which then activates glycogen phosphorylase;
8 the message is relayed in a **phosphorylation cascade**;
9 each kinase activates a large number of the next (relay) molecule, resulting the the signal being **amplified**/ a single glucagon molecule causes a strong response inside the cell

Cellular Response

10 glycogen phosphorylase catalyses the conversion of glycogen (large number) to glucose-1-phosphate, restoring blood glucose to the norm.

Other cellular responses [**max 1 point]**

11 Inhibition of enzymes needed for the formation of glycogen from glucose
12 Increase breakdown of lipids and proteins and conversion of other non-carbohydrate sources (eg. lactate) to glucose in liver cells (gluconeogenesis)
13 Increases the use of fatty acids (instead of glucose as the main fuel) in respiration.

(c) Describe the differences between the cell signalling systems of insulin and glucagon. [5]

	Insulin	**Glucagon**
Type of receptor	Insulin receptors belong to the family of receptor tyrosine kinases	Glucagon receptors belong to the family of G-protein coupled receptor
Ligand-receptor interaction	Dimerization of polypeptide involved	No dimerization of polypeptide
Ligand-receptor interaction	Activated receptors results in the phosphorylation of tyrosine residues on the other receptor **by tyrosine kinase** domain in a process known as auto-crossphosphorylation	Activated receptor results in the **binding of G protein**, which **exchanges** GDP for GTP and becomes **activated**
Signal transduction	Relay proteins **bind to** phosphorylated **tyrosine** residues, becomes **activated**	Activated G protein **binds** to and **activates adenylyl cyclase,** which catalyses the conversion of ATP to cAMP
Cellular response	**Movement** of vesicles containing glucose transporters (Glut4 proteins) towards and **fusion** with the cell surface membrane, **incorporating** glucose transporters on the cell surface membrane, leading to increased uptake/permeability of glucose	Glycogen phosphorylase catalyses the conversion of glycogen (large number) to glucose-1-phosphate, restoring blood glucose to the norm.
Cellular response (Max 1)	▪ Increases **conversion of glucose to glycogen in liver for storage (glycogenesis)** / Decreases **breakdown of glycogen in liver and muscle (glycogenolysis)** ▪ Decreases **breakdown of lipids and proteins and conversion of other non-carbohydrate sources to glucose in liver cells (gluconeogenesis)**	▪ Decrease conversion of glucose to glycogen (glycogenesis) / Increase **breakdown of glycogen (stored in liver) to glucose (glycogenolysis)** ▪ Increase **breakdown of lipids and proteins and conversion of other non-carbohydrate sources (eg. lactate) to glucose in liver cells**

	▪ Increases **the rate of <u>glucose oxidation</u> and synthesis of ATP**	**(<u>gluconeogenesis</u>)** ▪ Increases **the use of fatty acids (instead of glucose as the main fuel) in respiration.**
Other differences in Signal transduction (Max 1)	▪ No involvement of adenylyl cyclase ▪ No involvement of cAMP as second messenger	▪ Involvement of adenylyl cyclase ▪ Involvement of cAMP as second messenger

9

(a) Describe, using examples, how homology supports Darwin's theory of natural selection. [10]

1 Homology provide evidences supporting evolution as a remodelling process in which characteristics present in the ancestral organisms are altered by natural selection in its descendants over time

2 As they face different environmental conditions

Anatomical homology

3 E.g. The forelimbs of all mammals, including humans, cats, whales and bats show the same arrangement of bones from the shoulder to the tips of the digits, even though these appendages can have very different functions – lifting, walking, swimming and flying.

/ Vestigial structures **have no apparent function but resemble structures their presumed ancestors had.**
+ 1 e.g. skeletons of some snakes/whales **retain vestigial structures of the pelvis and** leg bones **of walking ancestors/** manatees **have** vestigial fingernails **on their fins (which evolved from legs)/**humans **possess a complete set of** muscles for wriggling of ears **of no apparent function**

4 Homologous body structures between organisms with **different functions i**mplies on evolutionary divergence from a common ancestor with descent with modification;

Embryological homology

5 **Comparative embryology**: **comparison** of **early developmental stages** of organisms revealed that **embryos** from different species **retain similarities during their development.**

6 Some of these **shared characteristics** in the embryo **might be lost when the adult is fully developed/**not visible in adult organisms.

7 E.g. At some point in their early development, all vertebrate embryos possess a tail and pharyngeal pouches which later develop gill slits in fishes and parts of ears and throat in humans.

8 Such similarity/homology in embryo development in very different organisms suggests inheritance from **common ancestor**, where these similar structures later develop into homologous structures with very **different functions**, implying on **descent with modification.**

Molecular homology

9 Comparison between organisms at the molecular level (RNA and proteins) revealed similarities, suggesting inheritance from a **common ancestor**

10 **2 species that are more closely related should share a greater similarity in their DNA.**

11 As the descendants of same species evolve independently, more and more differences (mutations) are accumulated in their DNA.

12 There could be subsequent morphological diversity between organisms as a result of divergent evolution, indicating of descent with modification.

.

9b) Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [10]

Hb^SHb^S individuals

1 Individual homozygous for Hb^S / genotype Hb^SHb^S suffer from sickle cell anaemia
2 the normal haemoglobin (haemoglobin A) in red blood cells is replaced entirely by abnormal haemoglobin (haemoglobin S).
3 Sickled red blood cells tend to stick together and obstruct blood flow in capillaries, depriving multiple organs of oxygen, resulting in organ damage / Sickled red blood cells rupture easily, resulting in anaemia and fatigue.
4 May result in early death. Therefore, there is a higher potential for the removal of the Hb^S allele from the gene pool.

Hb^AHb^A individuals

5 Hb^AHb^A individuals are at greater risk of dying of malaria compared to heterozygotes. Therefore, there is a higher potential for the removal of the Hb^A allele from gene pool.

Hb^AHb^S individuals

6 Individual who are heterozygous / genotype Hb^AHb^S suffer from sickle cell trait
7 produce both normal and abnormal haemoglobin
8 Hb^AHb^S individuals are a selective advantage and are able to survive and reproduce in malaria-infected areas.
9 (This condition is known as heterozygote advantage as) it preserves the Hb^S allele in the population
10 A mosquito transmits *Plasmodium*, the malaria parasite to humans. The malaria parasite spends part of its life cycle in red blood cells.
11 When malaria parasite invade the bloodstream, the red blood cell that contains haemoglobin S is sickled-shaped and is quickly destroyed by the body, trapping the parasites within the sickle cell and stopping the infection.
12 The slowdown in blood flow in sufferers of sickle cell trait also hampered the parasite’s ability to travel and rapidly infect new cells.